POWER SERIES, QUADRATIC SEQUENCES AND CANCELLATION

Euler's pentagonal formula

Why do the powers $0, 1, 2, 5, 7, 12, 15, \ldots$ survive an infinite product?

1 / READ MULTIPLICATION AS A CHOICE

An exponent records a total; its coefficient records a signed count

$$F(x) = \prod_{n=1}^{\infty}(1 - x^n) = 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + \cdots .$$

From each factor choose 1 or $-x^n$. Choosing distinct integers $a_1, \ldots, a_r$ contributes $(-1)^r x^{a_1+\cdots+a_r}$. A **partition** is a sum in which order is ignored. Thus the coefficient of x^N counts partitions of N into distinct parts, with a plus for an even number of parts and a minus for an odd number.

2 / THE NATURE OF THE INDEX SEQUENCE

One quadratic, read at positive and negative integers

Set $g(k) = \dfrac{k(3k-1)}{2}$. Read k in the order $0, 1, -1, 2, -2, 3, -3, \ldots$; the resulting $g(k)$ are the **generalised pentagonal numbers**.

k	0	1	−1	2	−2	3	−3	4	−4
$g(k)$	0	1	2	5	7	12	15	22	26
$(-1)^k$	+	−	−	+	+	−	−	+	+

For positive k, the ordinary pentagonal numbers are $1, 5, 12, 22, \ldots$, built by adding $1, 4, 7, 10, \ldots$ from zero:

$$g(k) = 1 + 4 + \cdots + (3k-2), \qquad g(-k) = g(k) + k \quad (k \geq 1).$$

Each branch has second difference 3. Their merged gaps $1, 1, 3, 2, 5, 3, 7, 4, \ldots$ alternate odd numbers with $1, 2, 3, 4, \ldots$. Also $24g(k) + 1 = (6k-1)^2$: a non-negative integer N survives exactly when $24N + 1$ is a square. After the constant, signs come in pairs.

3 / DERIVE THE CANCELLATION: FRANKLIN'S PAIRING

Move dots, preserve the total, change the sign

Draw distinct parts as left-aligned rows, longest first. Let s be the shortest row's length, and let d count the initial run of row lengths decreasing by exactly 1.

If $s \leq d$: remove the bottom row and add one of its dots to each of the first s rows.

If $s > d$: remove one end-dot from each of the first d rows and make these dots a new bottom row of length d.

Whenever valid, these moves undo each other. They keep the total N and change the number of rows by 1: the paired terms cancel. Such a self-reversing pairing is called an **involution**.

A cancelling pair at total 6



$(4, 2)$ $+x^6$ ⟷ $(3, 2, 1)$ $-x^6$

The other pair is $(6) \longleftrightarrow (5, 1)$.

All contributions to x^6 cancel: $-1 + 1 + 1 - 1 = 0$.

4 / FIND THE SHAPES THAT CANNOT BE PAIRED

The exceptional staircases explain the surviving powers

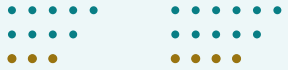
With k rows, the first move fails only at $s = d = k$ (a target row was removed); the second only at $d = k$, $s = k + 1$ (two rows become equal). Their consecutive row lengths give

$$k + (k+1) + \cdots + (2k-1) = \frac{k(3k-1)}{2},$$

$$(k+1) + (k+2) + \cdots + 2k = \frac{k(3k+1)}{2}.$$

Both have k parts, hence sign $(-1)^k$.

Two survivors with $k = 3$ rows



$5 + 4 + 3 = 12$ $-x^{12}$ $\qquad$ $6 + 5 + 4 = 15$ $-x^{15}$

Bottom rows have lengths 3 and 4. With all three rows consecutive, either move fails.

5 / EULER'S PENTAGONAL NUMBER THEOREM

$$\prod_{n=1}^{\infty}(1 - x^n) = \sum_{k \in \mathbb{Z}} (-1)^k x^{k(3k-1)/2}$$

The empty choice supplies 1. Every other term cancels except the two k-row staircases above.

Formal identity: the coefficient of x^N uses only the first N factors. The product and series also converge for $|x| < 1$.

Sources. NIST DLMF 27.14.4–5; Igor Pak, *The Nature of Partition Bijections I*, §1.1; York University notes on Franklin's proof.

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